

Flagella-Cascade Investigation — Lab Update

Notes

Prep notes for research-update presentation · Maya Abdalla · Aim 2 (Kalir & Alon SUM-gate) + Aim 2B (NFsim rule-based assembly)

Timeline since the prelim exam

Jun 20 — Investigation kickoff. Ported the Kalir & Alon transcriptional SUM-gate cascade (Aim 2) from the vEcoli `biofilm` branch into v2ecoli. Brought in NFsim (Aim 2B) from `pbg-nfsim` the same day. Split into studies, ran initial multi-seed ensembles; found and fixed a full-mode calibration issue that had reversed the headline flagella-count result.

Jul 4–25 — Infrastructure and rebasing: kept pace with a codebase-wide `baseline→ecoli_baseline` rename, registered `flagella_nfsim_assembly` as its own composite, added a SQLite-emitting study runner, restructured into 5 studies.

Jul 27 — Responded to lab review feedback ("Notes on v2ecoli"): config/run docs, a workflow-report bug, reconstruction audit.

Aug 7–8 — Real stoichiometry fixes, a FliT checkpoint mechanism added, an fliC synthesis fix, and the first documented "WCM division limitation." A reduced-target diagnostic (target=500) revealed a second supply bottleneck distinct from the checkpoint.

Aug 11 — Removed the FliT:FlhDC checkpoint entirely, after a literature check (Albanna et al. 2018 found no significant Δ -fliT phenotype in the K-12 strain this model uses). Fixed the export-apparatus/MS-ring assembly hierarchy and a same-tick SSA/Step race condition.

Aug 12 — NFsim wiring plan, steps 1–2: real bulk IDs, real ambient seeding, ported the NFsim model into v2ecoli, fixed a hook-completion chunking bug (became a PR to the upstream `pbg-nfsim` repo).

Aug 13–19 — The bulk of this update (details below): NFsim wired in as a real simulation Step; a cascade of root-caused infrastructure and biology findings; a real missing chaperone mechanism identified from literature, implemented, and tested.

Part 1 — Infrastructure: NFsim wired into the real whole-cell model

`flagella_nfsim_complexation.py`, a new Step, replaces the custom deterministic motor-switch/export-apparatus/motor-complex/nucleation Steps inside the actual whole-cell composite. Regulation (the Kalir & Alon SUM-gate) and secretion are untouched and reused as-is — only the complexation mechanism itself changed, from hand-written deterministic code to rule-based stochastic simulation (NFsim/BioNetGen).

Real bugs found and fixed along the way

- A hook-nucleation-rate bug caused 1,226 stuck partial assembly scaffolds in a single interval — traced, fixed, verified (0 stuck afterward).
- Long test runs kept getting silently killed; root-caused to session/task-tracking teardown (not, as first suspected, the machine sleeping) — fixed by fully detaching long runs from the tool session.
- A hard crash on an 8-hour run (NFsim hit its own internal molecule-count limit) was proven — by direct measurement, not guessing — to be a real consequence of running many hours with cell division disabled (no dilution sink for unconsumed structural monomers), not a coupling bug. This reframed how long any single diagnostic run should target.
- **A real division-state bug:** two new state variables the NFsim Step introduced (in-progress partial assemblies, internal-only species tallies) were silently reset to empty at every cell division — they were never actually included in the division machinery at all. Fixed with a biologically-grounded binomial-per-structure divider, verified on real accumulated simulation state.

Part 2 — The science: why is flagellar completion rare?

A complete causal chain, each link independently verified against real data or literature:

2.1 — Nucleation is genuinely rare, by design

C-ring nucleation is calibrated to a real literature rate (Sisti et al. 2017, back-calculated from measured *E. coli* flagella-count-vs-growth-rate data). Fine-grained (10-second resolution) sampling confirmed nucleation events are brief and rare — invisible at coarser sampling, present as 1–2-count blips only at fine resolution.

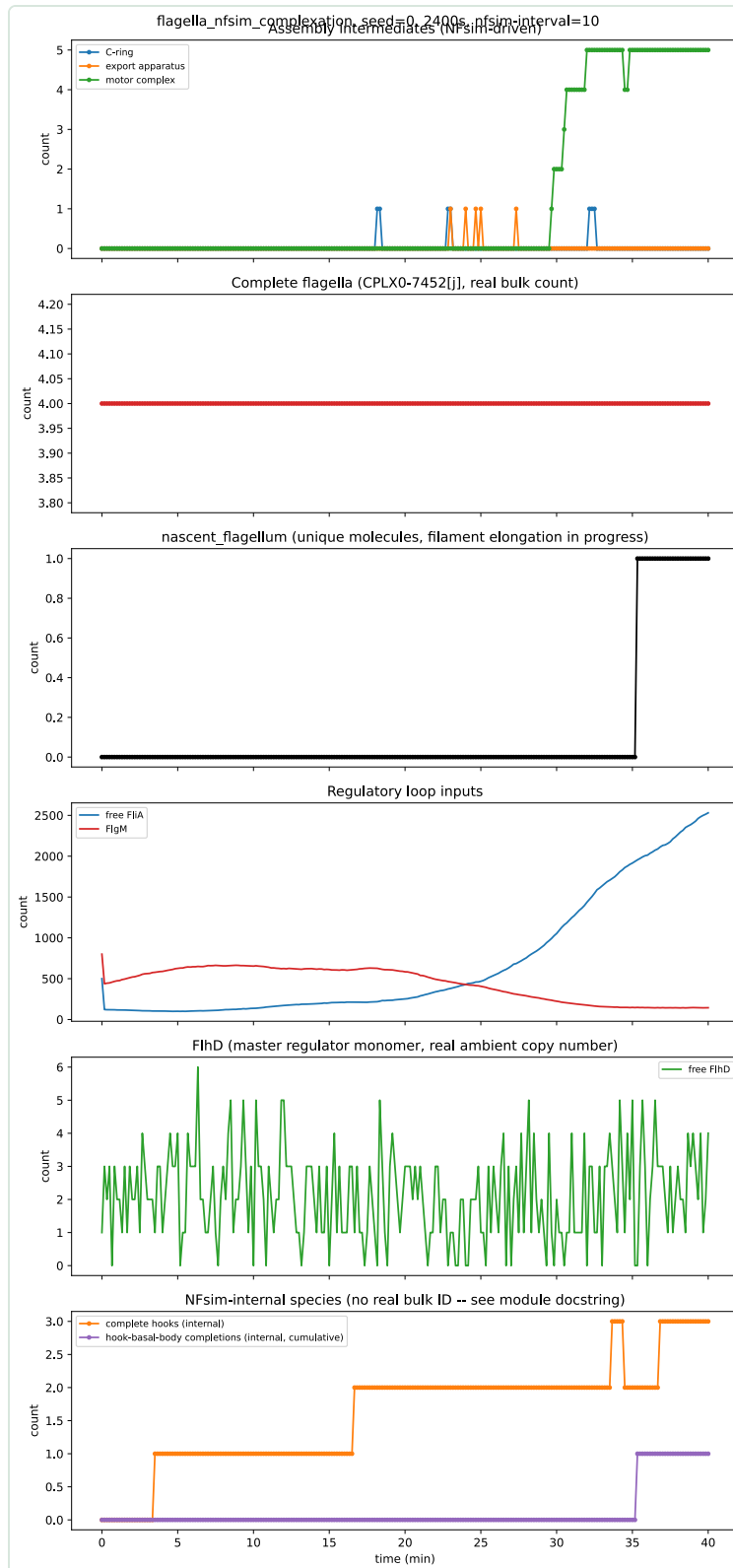


Figure. 10-second-resolution single-cell trace. C-ring and export-apparatus intermediates (top-left) only register as brief, isolated blips — confirming nucleation is a rare, real event rather than a sustained population, matching the Sisti et al. 2017 calibration.

2.2 — Filament elongation needs ~5,000 flagellin (FliC) subunits, and supply is doubly constrained

Elongation kinetics (calibrated from Renault et al. 2017 filament-growth data) predict a theoretical minimum of **~9.7 minutes** to build one filament if FliC were unlimited. In practice it takes much longer:

- **Synthesis is capped.** The existing fliC transcription override (10×, from earlier work) was already documented as the max defensible bump — going further would take one gene to ~13.5% of the cell's entire shared transcription budget. Even at 10×, real synthesis is still ~65× below the literature-derived peak demand.
- **Degradation is fast — and real, not assumed.** Queried live from the model: FliC's degradation rate is **$k = 0.00156/\text{s} \rightarrow 7.4\text{-minute half-life}$** , traced to its exact literature source (Nagar et al. 2022, pulsed-SILAC proteomics). Cross-checked against the full model proteome: the **99.7th percentile fastest-degrading protein** in the entire model — ~81× faster than its own flagellar-pathway siblings (FlgE, FliD, FlgK, FlgL, all ~10-hour half-life). Confirmed via the original paper's methodology that this measurement is properly separated from growth dilution — a real proteolysis rate, not a modeling artifact.

Net effect, observed directly: free FliC crashes from an ambient ~52,000 molecules down to ~1,000–2,500 during active elongation — a real, measured supply crisis.

2.3 — The missing mechanism: FliS, a real chaperone the model didn't represent

Real bacteria solve exactly this problem with FliS, a dedicated export chaperone that binds newly-made flagellin and shields it from cytoplasmic degradation during the handoff to the export apparatus. FliS is a real gene already in the model — it was simply never functionally wired into the assembly pathway.

Implemented as a real reaction, not a tuned parameter: $\text{FliS} + \text{FliC} \rightleftharpoons \text{FliS:FliC}$ complex, using real, sourced binding kinetics ($K_a = 1.9 \times 10^7 \text{ M}^{-1}$, 1:1 stoichiometry — Muskotal et al. 2006, isothermal titration calorimetry). Mechanistically clean: a complex is a structurally different species from a free monomer, so it's automatically exempt from the degradation process — verified in the simulation code before implementing.

The full whole-cell parameter fit was rebuilt with this reaction added (a multi-hour, from-scratch reconstruction) — completed cleanly, no numerical instability. The protected complex forms immediately at runtime and saturates FliS's own real abundance (~2,000 molecules protected at a time, scaling with population size), exactly as predicted before the rebuild.

2.4 — Honest result: it helped, but didn't fully solve it

Comparison runs (before/after the FliS fix, same conditions): completions happened roughly **2x faster** in the best comparison, and a filament that had stalled at 35% progress before the fix reached 89% in the equivalent window afterward. **But** the underlying pattern — average flagella-per-cell declining every generation as division dilutes faster than synthesis replenishes — persisted even with the fix, just from a higher starting point. A real reproducibility caveat also surfaced: NFsim's own internal random seed isn't tied to the model's overall seed, so identical launches can produce meaningfully different trajectories — meaning single-run comparisons this session are suggestive, not statistically confirmed.

Simulation result — population-level test, real division, FliS-protected

Full comprehensive panel set (regulatory chain, FliS protection, assembly cascade, per-filament trajectories), summed across every live cell in an expanding population undergoing real cell division:

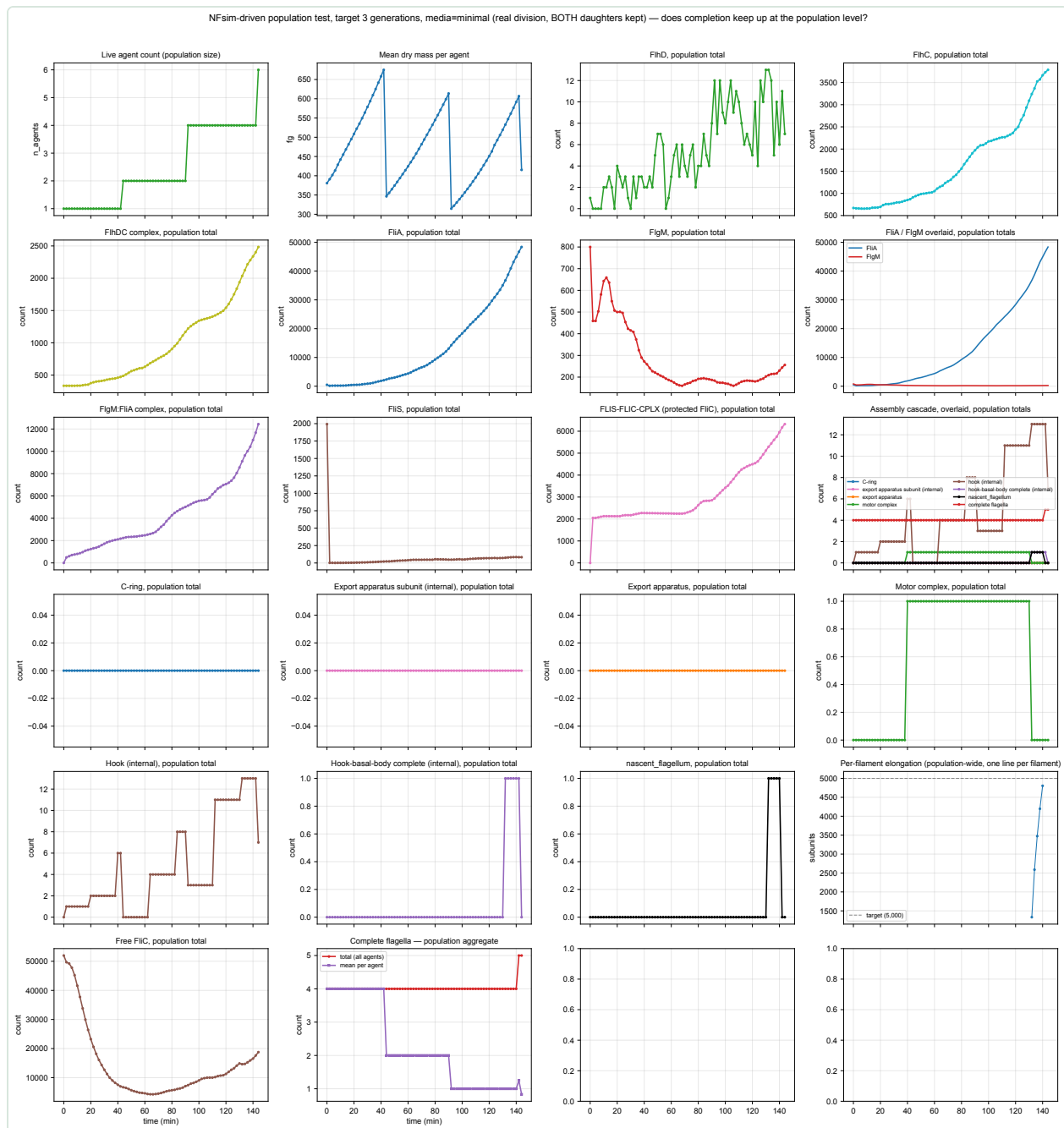


Figure. Population-level test, 3 generations, real division (both daughters kept at each division), FliS-protection mechanism active. 22 panels covering the full regulatory-to-assembly cascade as population totals. Panel "Complete flagella — population aggregate" shows the key result: total flagella count (red) vs. mean per cell (purple) as the population divides.

Where this leaves things

Real *E. coli* reliably builds flagella. This model, even after adding a literature-grounded missing mechanism, still shows fragile, high-variance completion. That gap is itself

informative — it suggests either another real mechanism is still missing/underrepresented, or a rate constant somewhere in the existing chain still needs recalibration. Framed honestly: a rigorous causal chain traced end-to-end, a partial real fix implemented and tested, and a clear, specific list of what's still open — not "problem solved."

Explicitly deferred, not forgotten: true spatial/compartment modeling, which would let FliS protection work exactly as real biology does (protecting a cytoplasmic pre-export pool specifically) rather than as a well-mixed approximation — tied to future spatiality work on the model generally.

Key numbers

Quantity	Value	Source
Filament target length	5,000 subunits	this model's calibration
Theoretical min. elongation time (unlimited FliC)	~9.7 min	derived from Renault et al. 2017 kinetics
fliC synthesis, post-10× override	~7 molecules/s	vs. ~46/s literature peak demand
FliC degradation half-life	7.4 min ($k=0.00156/s$)	Nagar et al. 2022, pulsed-SILAC
FliC degradation percentile (whole proteome)	99.7th fastest	measured directly from model
FliS:FliC binding affinity	$K_a = 1.9 \times 10^7 \text{ M}^{-1}$ ($K_d \approx 53 \text{ nM}$)	Muskotal et al. 2006
FliS real ambient abundance	~2,000–2,100 molecules/cell	this model's parameter fit
Nucleation rate calibration source	Sisti et al. 2017	<i>Sci Rep</i> 7:41189

Sources cited

Sisti F, et al. (2017). Growth rate control of flagellar assembly in *E. coli* strain RP437. *Sci Rep* 7:41189.
Renault TT, et al. (2017). Bacterial flagella grow through an injection-diffusion mechanism. *eLife* 6:e23136.
Nagar T, et al. (2022). Pulsed-SILAC protein half-life measurements in *E. coli*.
Muskotal A, et al. (2006). Interaction of FliS flagellar chaperone with flagellin. *FEBS Lett*.
Evdokimov AG, et al. (2003). Similar modes of polypeptide recognition by export chaperones in flagellar

biosynthesis and type III secretion.

"How bacteria maintain location and number of flagella?" — FEMS Microbiology Reviews (peritrichous flagellar inheritance).

Albanna A, et al. (2018). FliT phenotype comparison, *E. coli* K-12 vs. *Salmonella*.